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TopoCL: Topological Contrastive Learning for Time Series

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***Abstract*—Universal time-series representation learning is challenging but valuable in real-world applications such as classification, anomaly detection, and forecasting. Recently, contrastive learning (CL) has been actively explored to tackle time-series representation. However, a key challenge is that the data augmentation process in CL can distort seasonal patterns or temporal dependencies, inevitably leading to a loss of semantic information. To address this challenge, we propose topological CL for time series (TopoCL). TopoCL mitigates such information loss by incorporating persistent homology, which captures the topological characteristics of data that remain invariant under transformations. In this article, we treat the temporal and topological properties of time-series data as distinct modalities. Specifically, we compute persistent homology to construct topological features of time-series data, representing them in persistence diagrams (PDs). We then design a neural network to encode these persistent diagrams. Our approach jointly optimizes CL within the time modality and time–topology correspondence, promoting a comprehensive understanding of both temporal semantics and topological properties of time series. We conduct extensive experiments on four downstream tasks—classification, anomaly detection, forecasting, and transfer learning. The results demonstrate that TopoCL achieves state-of-the-art performance.**

***Index Terms*—Anomaly detection, classification, contrastive learning (CL), forecasting, time-series modeling, topological data analysis (TDA).**

I. Introduction

TIME-SERIES analysis plays a crucial role in numerous real-world applications such as weather, economics, and transportation. However, analyzing time-series data are challenging due to its incompleteness, vulnerability to noise, and complexity. In addition, employing deep learning models for time-series analysis introduces further obstacles, as it necessitates well-labeled data. Labeling time-series data is time-consuming and requires a high level of expertise, creating significant bottlenecks in the analysis process. In recent years, contrastive learning (CL) has become increasingly popular in graph [1], computer vision (CV) [2], [3], natural language processing (NLP) [4], and recommendation system [5], [6], mainly due to its ability to train models without explicit labels. Accordingly, ongoing research is focused on applying CL to time-series data for various downstream applications, including anomaly detection [7], [8], forecasting [9], and classification [10].

In general, CL leverages data augmentation to create positive and negative pairs for training, thereby creating a contrast between similar and dissimilar samples by exploring different transformations and variations of the data. Several augmentation techniques for time-series have been proposed to learn robust and discriminative representations. For example, *permutation* reorders segments within a time series to produce a new sequence [10], [11]. However, using *permutation* on time-series data might introduce inappropriate inductive bias, as it can distort the autocorrelated nature of time-series data. Techniques borrowed from CV, such as *flipping*, which flips the signs of the original time series, may misrepresent the trend or seasonal patterns. Moreover, these distortions raise concerns about whether the augmented time series are, indeed, positive samples of the original series. Consequently, applying data augmentation may inevitably risk information loss, making it difficult to capture semantically meaningful representations. Therefore, it is necessary to find ways that compensate such potential information loss.

Recently, topological data analysis (TDA) has emerged as a subfield of algebraic topology focused on capturing the global *shape* of data, where *shape* broadly refers to data properties that remain invariant under transformations [12]. Such shape patterns or topological features include connected components, loops, and voids. Topological features are obtained by computing persistent homology and are represented as a multiset in persistence diagrams (PDs) [13]. These diagrams encapsulate persistent homology across different scales of data. The overall structure and distribution of sets in the diagram provide insights into the topological invariants of the data. By incorporating such topological information, we can capture the essential structural properties of the time series that remain invariant under various transformations. Therefore, integrating topological information can help compensate for potential information loss due to data augmentation, reduce inductive bias, and lead to more robust representations.

Fueled by the advancements in both CL and TDA, we harness the strengths of the two methods and introduce topological CL for time series (TopoCL), a simple yet effective cross-modal CL approach for universal time-series representations. Our method integrates and leverages information from the time and topology modalities. To start, we construct

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topological characteristics from time series data. We apply delay embedding to time-series data, compute persistent homology, and design a simple topological module to effectively encode topological information. Our approach then focuses on a joint objective: ensuring that the augmented versions of the same time series are closely embedded together in the feature space while preserving the correspondence between time and topology. The joint learning objective promotes a comprehensive understanding of temporal semantics and topological properties of time series and improves robustness to data augmentations. We perform extensive experiments on time-series forecasting, anomaly detection, classification, and transfer learning tasks. The experimental results demonstrate that the learned representations of TopoCL are effective. The contributions of this work are as follows.

1) We demonstrate that incorporating topological properties is effective for capturing robust and discriminative representations of time-series data.
2) We design a novel framework for learning representations of time-series data that takes into account their topology. To the best of our knowledge, our approach is the first to combine persistent homology with CL in the context of time-series representation learning.
3) We conduct extensive experiments across four downstream tasks: time-series classification, anomaly detection, forecasting, and transfer learning on diverse datasets. The experimental results demonstrate the effectiveness of the proposed model.

The remainder of this article is organized as follows. Section II presents a literature review on TDA and time-series CL. Section III provides preliminaries. The proposed framework is described in Section IV. Section V provides the experimental results of the proposed model. Section VI presents the additional analysis. Finally, Section VII presents the conclusion and future work.

II. Related Works

A. Topological Data Analysis

TDA is an emerging field that utilizes abstract algebra to uncover the intrinsic shape of the data. A fundamental concept in TDA is PD, which helps to understand the topological structure of data. By computing persistent homology across scales, such topological structure is represented as a multiset of points. Directly using PD in machine learning and deep learning models is challenging, since these models typically work with Hilbert space. To overcome this, several research efforts have been made to convert PD into a vector format. For instance, persistence landscapes transform the multiset of points in a PD into a collection of piecewise-linear functions [14]. Adams et al. [15] proposed persistence images, which convert PD into a fixed-size representation. In addition, various kernel functions have been proposed to handle PD, such as geodesic topological kernel [16] and persistence landscape-based kernels [17], [18].

Recently, the integration of TDA into neural networks has been explored, leading to the development of various topological layers for machine learning applications. The first approach to input a PD into a neural network architecture was presented by Hofer et al. [19]. Carrière et al. [20] proposed PersLay, which incorporates PD for graph classification. Moor et al. [21] proposed a topological autoencoder preserving topological structures of the input space in latent representations. Kim et al. [22] proposed PLlay, a neural network layer for learning embeddings from persistence landscapes. These approaches demonstrated that incorporating persistent homology with deep learning has the potential to offer a more comprehensive understanding of data.

B. Time-Series CL

Similar to the advancements of CL in the fields of CV and NLP, numerous studies incorporate CL into time-series analysis. TS-TCC [10] employed both weak and strong augmentations to time-series data and utilized contextual contrasting to learn transformation-invariant representations. Mixing-up [23] generated an augmented sample by mixing two data samples and proposed a pretext task method for predicting the mixing proportion. Inspired by great success in masked modeling in NLP and CV, several works adopted this approach for time series. TS2vec [7] used masking and cropping for data augmentation, and proposed a hierarchical contrastive loss to learn scale-invariant representations. InfoTS [24] exploited a meta-learning framework to automatically select the augmentation method. SimMTM [25] generated multiple masked series and facilitated reconstruction by assembling complementary temporal variations from multiple masked series.

Recently, several studies have focused on the use of the frequency domain of time series data. CoST [9] proposed a disentangled seasonal-trend representation learning framework, incorporating contrastive loss in both the time and frequency domains. TF-C [26] proposed a novel contrastive loss to maintain consistency between frequency- and time-domain representations.

Gorade et al. [27] captured both low- and high-frequency features from the input signal and learns complementary features between them to learn robust representations. TimesURL [8] proposed a novel frequency-temporal-based augmentation method to maintain temporal dependencies of time series. However, it still remains unclear whether the proposed methodologies effectively mitigate the inevitable information loss resulting from data augmentation. To address this issue, we propose using persistent homology to capture the essential structural properties of time-series data, thereby mitigating such information loss.

III. Preliminary

In this section, we give the formal definitions of the major terminologies of TDA used in this study.

A. Delay Embedding

Takens's delay embedding [28] is a commonly used technique for converting time-series data into point cloud representations. Given an ith time series $x_i = \{x_{i,1}, x_{i,2}, \ldots, x_{i,T}\}$, one can extract a sequence of vectors of the form

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$[x_{i,t}, x_{i,t+\gamma}, \ldots, x_{i,t+(m-1)\gamma}] \in \mathbb{R}^m$, where m is the embedding size and γ is the time delay.

B. Simplicial Complex

In the algebraic topology, a simplex is a geometric object that generalizes the notion of a triangle or a tetrahedron to higher dimensions. Specifically, a k–simplex is defined as a convex hull spanned by $k+1$ points $v \in \mathbb{R}^m$ that are affinely independent. Then, a simplicial complex $\mathcal{K}$ is a finite collection of simplices such that following hold.

1) If $\sigma \in \mathcal{K}$, every subset of σ (i.e., every face of σ) is an also element of $\mathcal{K}$.
2) If two simplices $\sigma_1, \sigma_2 \in \mathcal{K}$ intersects, their intersection is a face of both σ_1 and σ_2.

C. Vietoris–Rips Complex

To construct a simplicial complex, we used the Vietoris–Rips complex [29]. Let (X, d) be a finite metric space equipped with a distance function d. The Vietoris–Rips complex $VR(X, r)$ is the abstract simplicial complex whose simplices are subsets of X with diameter (i.e., the maximum pairwise distance between points) at most $r \geq 0$. That is, a subset S of X is a simplex in $VR(X, r)$ if and only if the distance $d(x, y) \leq r$ for any two points $x, y \in S$. The 0-simplices of $VR(X, r)$ are the points of X, and the k-simplices are the subsets of X of cardinality $k+1$ that can be realized as the union of $k+1$ closed balls of radius r with nonempty pairwise intersections.

D. Homology

1) Simplicial Homology: Given a simplicial complex $\mathcal{K}$, a weighted sum of n-simplices with coefficients from $\mathbb{Z}/\mathbb{Z}_2$-defines an n-chain on $\mathcal{K}$. Then, the space of n-chains on $\mathcal{K}$, represented as $C_n(\mathcal{K})$, is defined as the vector space spanned by n-simplicies of $\mathcal{K}$ over $\mathbb{Z}/\mathbb{Z}_2$. The modules $C_0, C_1, \ldots,$ are connected by boundary operators $\partial_n : C_n(\mathcal{K}) \rightarrow C_{n-1}(\mathcal{K})$. The boundary operator for a simplex $\sigma = [v_0, \ldots, v_n]$ is defined as follows:

$$\partial_n(\sigma) = \sum_{i=0}^{n} [v_0, \ldots, \hat{v}_i, \ldots, v_n] \tag{1}$$

where $[v_0, \ldots, \hat{v}_i, \ldots, v_n]$ denotes the $(n-1)$ simplex spanned by all the vertices except v_i. It is easy to see that $im(\partial_{n+1})$ is contained in $ker(\partial_n)$ for each n. The nth homology group of $\mathcal{K}$ is defined as follows:

$$H_n(\mathcal{K}) = ker(\partial_n) / im(\partial_{n+1}). \tag{2}$$

The nth Betti number of $\mathcal{K}$, denoted by $\beta_n(\mathcal{K})$, is the dimension of the nth homology group of $\mathcal{K}$, which can be computed as the difference between the kernel dimension of ∂_n and the image dimension of ∂_{n+1}. In simpler terms, the nth Betti number indicates the number of n-dimensional voids that exist in the simplicial complex. For example, β_0 indicates the number of connected components, while β_1 represents the number of loops.

2) Persistent Homology: Persistent homology is a valuable technique in TDA for understanding the shape of complex datasets. For a given simplicial complex K, let $(K)_{i=0}^m$ be a nested subcomplexes of K such that $\emptyset = K_0 \subseteq K_1 \subseteq \ldots \subseteq K_m = K$, called a *filtration*. The inclusion maps $K_i \hookrightarrow K_j$ defined by $f(x) = x$ induce $f_{i,j} : H_p(K_i) \rightarrow H_p(K_j)$ for all i, j with $1 \leq i \leq j \leq m$, for all p. Then, the sequence of homology groups is represented as follows:

$$0 = H_p(K_0) \rightarrow H_p(K_1) \rightarrow \ldots \rightarrow H_p(K_m) = H_p(K). \tag{3}$$

For given $i \leq j$ and p, we can consider the pth boundary maps $\partial_p^i : C_p(K_i) \rightarrow C_{p-1}(K_i)$ and $\partial_p^j : C_p(K_j) \rightarrow C_{p-1}(K_j)$. Since $K_i \subset K_j$, one can see $C_p(K_i)$ and $C_{p-1}(K_i)$ as subspaces of $C_p(K_j)$ and $C_{p-1}(K_j)$, respectively. Then, both $im(\partial_{p+1}^j) \subset C_p(K_j)$ and $ker(\partial_p^i) \subset C_p(K_i)$ can be considered as subspaces of $C_p(K_j)$. Then, pth persistent homology group is defined as follows:

$$H_p^{i,j} = ker\left(\partial_p^i\right) \Big/ \left(im\left(\partial_{p+1}^j\right) \cap ker\left(\partial_p^i\right)\right). \tag{4}$$

Note that an element of $ker(\partial_p^i)$ represents both an element of $H_p(K_i)$ and an element of $H_p(K_j)$. If it vanishes in $H_p^{i,j}$, then it must be in $im(\partial_{p+1}^j)$, hence it vanishes in $H_p(K_j)$ as well. In summary, the homology group $H_p^{i,j}$ consists of the pth homology classes of K_i that are still present at K_j, i.e., those not in the kernel of the map $f_{i,j}$.

The pth Betti number, which refers to the rank of the homology group, is defined as: $\beta_p^{i,j} := \text{rank} H_p^{i,j}$. By analyzing the sequence of Betti numbers, persistent homology enables the systematic measurement of topological features (e.g., holes and voids) in a dataset across multiple scales. Such topological information can be summarized as PD.

E. Persistence Diagram

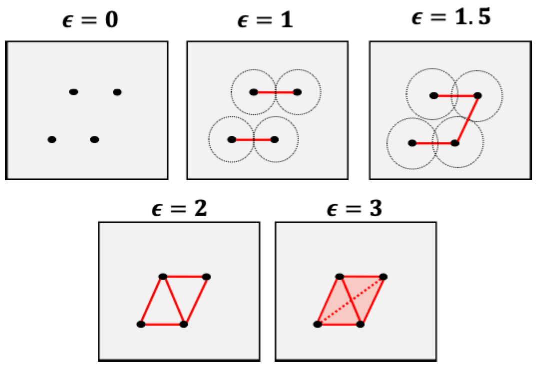
In TDA, PD is a useful tool for summarizing topological information from data across different scales. A PD captures topological features such as connected components and loops, showing when these features appear and disappear as the scale parameter changes. The pth PD Dgm_p is defined as a multiset collection of points (i, j) with $\mu_p^{i,j} = 1$, where $\mu_p^{i,j}$ is calculated as follows:

$$\mu_p^{i,j} = \left(\beta_p^{i,j-1} - \beta_p^{i,j}\right) - \left(\beta_p^{i-1,j-1} - \beta_p^{i-1,j}\right). \tag{5}$$

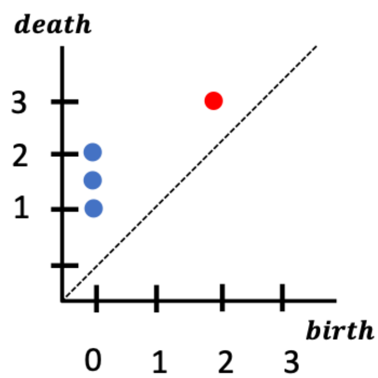
Each point $(a, b) \in \text{Dgm}_p$ corresponds to a homological feature that was born at scale a and dies at scale b. This homological feature has the persistence value of $b - a$. Fig. 1 illustrates an example of PD based on the Vietoris–Rips filtration. The diagram effectively summarizes evolving homological information with respect to ϵ. For instance, as ϵ increases from 0 to 1, the number of connected components decreases from 4 to 2, which is reflected as a point (0, 1) on the PD.

IV. Stability of Persistent Homology Under Various Data Augmentations

In Section III, we claim that the use of topological features as a robust auxiliary view in CL is stable under common data augmentations such as jittering, scaling, and shifting.



(a)



(b)

Fig. 1. Illustration of PD. (a) Vietoris–Rips filtration. (b) Corresponding PD.

We begin with the following well-established theory from persistent homology.

Theorem 1 (Rips Stability via Gromov–Hausdorff Distance [30]): Let (X, d_X) and (Y, d_Y) be compact metric spaces, and let F be a field. For every $k \geq 0$, the kth persistent homology modules of the Vietoris–Rips filtrations

$$\mathbf{H}_k(X; F) = \{H_k(\mathrm{VR}(X, r); F)\}_{r\geq 0}$$
$$\mathbf{H}_k(Y; F) = \{H_k(\mathrm{VR}(Y, r); F)\}_{r\geq 0}$$

are $2\, d_{\mathrm{GH}}(X, Y)$ interleaved

$$d_I(\mathbf{H}_k(X; F),\ \mathbf{H}_k(Y; F)) \ \leq\ 2\, d_{\mathrm{GH}}(X, Y).$$

In particular, for the associated PDs

$$d_B\big(\mathrm{Dgm}_k(X),\ \mathrm{Dgm}_k(Y)\big) \ \leq\ 2\, d_{\mathrm{GH}}(X, Y).$$

This theorem implies that if two point clouds $X, \tilde{X} \subset \mathbb{R}^d$ are close in Gromov–Hausdorff distance, then their Vietoris–Rips PDs are close in bottleneck distance. Therefore, when data augmentations cause only small geometric perturbations (i.e., when $d_{GH}(X, \tilde{X})$ is small), the corresponding Vietoris–Rips PDs remain stable.

To further demonstrate the robustness of persistent homology to practical data augmentations, we analyze four representative types of augmentations commonly applied to time series.

1) *Jittering:* Additive Gaussian noise applied to each time step
$$\tilde{x}_t = x_t + \epsilon_t, \quad \epsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma^2\right).$$

2) *Scaling:* Rescaling of the entire sequence
$$\tilde{x}_t = (1 + \epsilon)\ x_t, \quad \epsilon \sim \mathcal{N}\left(0, \sigma^2\right).$$

3) *Shifting:* Additive offset
$$\tilde{x}_t = x_t + \epsilon, \quad \epsilon \sim \mathcal{N}\left(0, \sigma^2\right).$$

4) *Rotation:* Sign inversion of the signal
$$\tilde{x}_t = -x_t.$$

In Sections IV-A–IV-D, we provide theoretical analyses showing that the PDs remain stable under each type of augmentation.

A. Stability Under Jittering

Let $\mathbf{x} = \{x_t\}_{t=1}^T$ be a real-valued time series and $\tilde{\mathbf{x}} = \{\tilde{x}_t\}_{t=1}^T$ its jittered version with $\tilde{x}_t = x_t + \epsilon_t$. Fix an embedding dimension m and delay τ, and form the delay-embedded point clouds

$$X = \left\{\mathbf{x}(i) = \left[x_i, x_{i+\tau}, \ldots, x_{i+(m-1)\tau}\right] \in \mathbb{R}^m\right\}_{i=1}^N$$
$$\tilde{X} = \left\{\tilde{\mathbf{x}}(i) = \left[\tilde{x}_i, \tilde{x}_{i+\tau}, \ldots, \tilde{x}_{i+(m-1)\tau}\right]\right\}_{i=1}^N \tag{6}$$

where $N := T - (m - 1)\tau$. For each i,

$$\|\mathbf{x}(i) - \tilde{\mathbf{x}}(i)\|_2^2 = \sum_{k=0}^{m-1} \epsilon_{i+k\tau}^2. \tag{7}$$

By the Laurent–Massart inequality [31], for any $s > 0$

$$\mathbb{P}\left(\|\mathbf{x}(i) - \tilde{\mathbf{x}}(i)\|_2^2 \leq \sigma^2\left(m + 2\sqrt{m\, s} + 2s\right)\right) \ \geq\ 1 - e^{-s}. \tag{8}$$

Since $d_H(X, \tilde{X}) \leq \max_{1\leq i\leq N} \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|_2$, we obtain

$$d_H\left(X, \tilde{X}\right) \ \leq\ \sigma\sqrt{m + 2\sqrt{m\, s} + 2s} \tag{9}$$

with probability at least $1 - N\ e^{-s}$.

Because $d_{\mathrm{GH}}(X, \tilde{X}) \leq d_H(X, \tilde{X})$ for subsets of the same metric space, the Vietoris–Rips stability theorem yields, for every $k \geq 0$

$$\begin{aligned} d_B\big(\mathrm{Dgm}_k(X),\ \mathrm{Dgm}_k(\tilde{X})\big) &\leq\ 2\, d_{\mathrm{GH}}\left(X, \tilde{X}\right) \\ &\leq\ 2\sigma\sqrt{m + 2\sqrt{m\, s} + 2s} \end{aligned} \tag{10}$$

with probability at least $1 - Ne^{-s}$.

B. Stability Under Scaling

For each i, we have $\tilde{\mathbf{x}}(i) = (1 + \epsilon)\mathbf{x}(i)$, hence

$$\begin{aligned} \|\mathbf{x}(i) - \tilde{\mathbf{x}}(i)\|_2 &= |\epsilon|\ \|\mathbf{x}(i)\|_2 \leq |\epsilon|\ M \\ M &:= \max_{1\leq i\leq N} \|\mathbf{x}(i)\|_2. \end{aligned} \tag{11}$$

Taking the canonical pairing $i \mapsto \tilde{\mathbf{x}}_i$ yields

$$d_H\left(X, \tilde{X}\right) \ \leq\ \max_{1\leq i\leq N} \|\mathbf{x}(i) - \tilde{\mathbf{x}}(i)\|_2 \ \leq\ |\epsilon|\, M. \tag{12}$$

Since $\epsilon \sim \mathcal{N}(0, \sigma^2)$, for any $s > 0$, Gaussian tail bounds yields

$$\mathbb{P}\left(|\epsilon| \leq \sigma\sqrt{2s}\right) \ \geq\ 1 - 2e^{-s}. \tag{13}$$

Consequently,

$$\mathbb{P}\left(d_H\left(X, \tilde{X}\right) \ \leq\ M\,\sigma\sqrt{2s}\right) \geq\ 1 - 2e^{-s} \tag{14}$$

$$\begin{aligned} d_B\left(\mathrm{Dgm}_k(X),\ \mathrm{Dgm}_k(\tilde{X})\right) &\leq\ 2\, d_{\mathrm{GH}}\left(X, \tilde{X}\right) \\ &\leq\ 2\, M\,\sigma\sqrt{2s} \end{aligned} \tag{15}$$

with probability at least $1 - 2e^{-s}$.

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C. Stability Under Rotation

For the sign-inverted time series $\tilde{\mathbf{x}}$, the delay-embedded point clouds X and $\tilde{X}$ are related by the reflection map $R(v) = -v$ for $v \in X$. Since R is an isometry, the metric spaces (X, d) and $(\tilde{X}, d)$ are distance-preserving isomorphic, and hence their Gromov–Hausdorff distance satisfies $d_{\mathrm{GH}}(X, \tilde{X}) = 0$.

Moreover, because the Vietoris–Rips complex depends only on pairwise distances, the filtrations $\{\mathrm{VR}(X, r)\}_{r\geq 0}$ and $\{\mathrm{VR}(\tilde{X}, r)\}_{r\geq 0}$ are isomorphic under S. Consequently, for every $k \geq 0$, the PDs coincide, i.e.,

$$\mathrm{Dgm}_k (X) = \mathrm{Dgm}_k \left(\tilde{X}\right) \tag{16}$$

and their bottleneck distance is 0. Therefore, Vietoris–Rips persistence is stable under sign inversion of the original time series.

D. Stability Under Shifting

For the shifted time series $\tilde{\mathbf{x}} = \{x_t + \varepsilon\}_{t=1}^{T}$, the delay-embedded point clouds X and $\tilde{X}$ are related by the translation map $T : X \to \tilde{X}$, defined by $T(u) = u + (\varepsilon, \ldots, \varepsilon)$ for $u \in X$. Since translation is an isometry, the metric spaces (X, d) and $(\tilde{X}, d)$ are isometric, and hence Vietoris–Rips persistence is also stable under shifting of the original time series.

These results collectively establish that PDs are robust to a wide class of common time-series augmentations. The key implication is that the distance between PDs is always bounded above by the amount of perturbation applied to the input.

V. Proposed TopoCL Framework

In this section, we present *TopoCL* framework.

A. Problem Formulation

Our goal is to train a neural network f_θ to transform each time series instance $x_i \in \mathbb{R}^{T\times C}$ in the time series set $X_{time} = \{x_1, x_2, \ldots, x_N\}$ into its representation r_i. Here, N denotes the number of samples, T denotes the number of timestamps, and C represents the number of variables. The representation r_i is expressed as $\{r_{i,1}, r_{i,2}, \ldots, r_{i,T}\}$, where $r_{i,t} \in \mathbb{R}^F$ is the embedding vector at time t, with F being the dimension of embedding vector.

B. Overview

The overall framework of TopoCL is shown in Fig. 2. Given X_{time}, we construct the topological modality X_{topo} by computing its PD. Then, X_{time} is fed into the *temporal module* to capture temporal dependencies. Concurrently, X_{topo} is processed by the *topological module* to learn the underlying topology of the data. Temporal features are learned using a time modality contrastive loss $\mathcal{L}_{time}$. In addition, an auxiliary contrastive objective $\mathcal{L}_{cross}$ is applied to align temporal features and topological features. The output of the temporal encoder f_θ^{time} is used for downstream applications.

C. PD Construction

To extract topological features from a time series, we first convert $x \in X_{time}$ into a point cloud format, enabling the construction of the Vietoris–Rips filtration. This involves applying delay embedding to the input sample x. Next, we compute PDs for H_0 and H_1 homology groups, summarizing the topological features of the data. For each channel c in time-series data, we obtain channel-specific 0-D and 1-D PDs $\mathrm{Dgm}^c(x)$. A composite diagram $\mathrm{Dgm}(x)$ is then created as $\mathrm{Dgm}(x) = \cup_c \mathrm{Dgm}^c(x)$. Next, points in $\mathrm{Dgm}(x)$ are mapped using $\delta : (a, b) \to (a, b, b - a)$, resulting in point cloud data $x^p \in \mathbb{R}^{M\times 3}$, where M is the maximum number of topological features. The set of topological feature is denoted by $X_{topo} = \{x_1^p, x_2^p, \ldots, x_N^p\}$. Further details of topological feature construction are provided in the Appendix.

D. Topological Feature Extraction

We design a topological feature extractor f_θ^{topo}, which directly processes unordered point sets x^p as inputs. Given that a point cloud is an unordered collection of points, a neural network capable of processing these M point sets needs to be invariant to $M!$ possible permutations of the input set's order during data feeding. Inspired by the works on PointNet [32] and Deep Sets [33], we apply a simple symmetric function to combine information from all the points in $x^p = \{p_1, p_2, \ldots, p_M\}$. Then, the model is unaffected by the order of input points, preserving invariance to input permutations.

Specifically, a series of shared multilayer perceptrons (MLPs) $\phi : \mathbb{R}^3 \to \mathbb{R}^H$ is applied to each topological feature $x_i^p \in X_{\mathrm{topo}}$ to obtain a set of encoded persistence points. We then aggregate the resulting embeddings via a permutation-invariant function ζ. Formally, the topological encoder is defined as follows:

$$f_{topo}^{\theta}\left(x_i^p\right) = \zeta\left(\left\{\phi\left(p_j\right)\right\}_{j=1}^{M}\right). \tag{17}$$

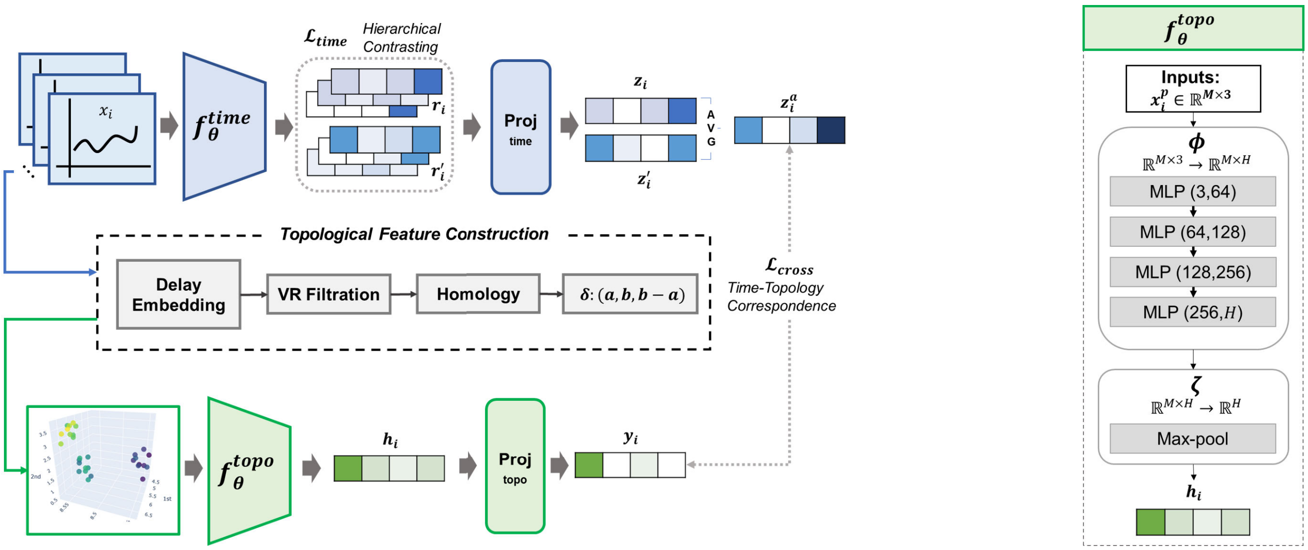
As illustrated in Fig. 2, we use a max-pooling layer as the permutation-invariant function ζ. Each point in x_i^p passes through the series of shared MLP with ReLU activation and max pooling aggregates all outputs into a topological representation vector $h \in \mathbb{R}^H$, where H is the dimension of the embedding vector. Since both the shared MLP ϕ and the max-pooling operator ζ are invariant to the order of the inputs, the overall function f_{topo}^{θ} is permutation invariant

$$f_{topo}^{\theta}\left(x_i^p\right) = f_{topo}^{\theta}\left(\pi\left(x_i^p\right)\right) \quad \forall \text{ permutations } \pi. \tag{18}$$

E. Temporal Feature Extraction

Capturing the temporal dependency is essential for time series representation learning. Given an input instance x_i, we construct augmented versions of it. Then, the time-series feature extractor f_θ^{time} maps them to a feature embedding space, resulting in embeddings r_i and r_i'.

To learn discriminative and robust temporal representations, we use temporalwise and instancewise contrastive losses, $\ell_{\mathrm{temp}}^{(i,t)}$



(a) (b)

Fig. 2. (a) Overall framework of TopoCL. TopoCL consists of two distinct modules: temporal module and topological module. The temporal module learns semantically meaningful temporal characteristics using a hierarchical contrastive loss, while the topological module facilitates time–topology correspondence through cross-modal contrastive loss. To achieve this, topological feature construction is performed in three steps: time delay embedding, PD calculation via Vietoris–Rips filtration, and point cloud transformation. Subsequently, a contrastive loss between the embeddings from the time and topology modalities is applied to ensure time–topology correspondence. TopoCL jointly optimizes the learning process through both intramodal and cross-modal correspondences. After pretraining, the embedding vectors obtained from f_θ^{time} are used for downstream applications. (b) Topological feature extractor f_θ^{topo}. The topological feature extractor comprises a series of MLPs with ReLU activations and a max-pooling layer.

and $\ell_{\text{inst}}^{(i,t)}$ [7]. For a given timestamp t, these two loss functions can be formulated as follows:

$$\ell_{\text{temp}}^{(i,t)} = -\log \frac{\exp\left(r_{i,t}\cdot r'_{i,t}\right)}{\sum_{t'}\left(\exp\left(r_{i,t}\cdot r'_{i,t}\right) + \mathbb{1}_{[t\neq t']}\exp\left(r_{i,t}\cdot r_{i,t}\right)\right)} \tag{19}$$

$$\ell_{\text{inst}}^{(i,t)} = -\log \frac{\exp\left(r_{i,t}\cdot r'_{i,t}\right)}{\sum_{j}\left(\exp\left(r_{i,t}\cdot r'_{j,t}\right) + \mathbb{1}_{[i\neq j]}\exp\left(r_{i,t}\cdot r_{j,t}\right)\right)}. \tag{20}$$

Time modality contrastive loss function L_{time} for a mini-batch B is defined as follows:

$$\mathcal{L}_{\text{time}} = \frac{1}{BT}\sum_i\sum_t\left(\ell_{\text{temp}}^{(i,t)} + \ell_{\text{inst}}^{(i,t)}\right). \tag{21}$$

Next, we apply the hierarchical contrasting method proposed by Yue et al. [7]. This involves performing max pooling on $r_{i,t}$ and $r'_{i,t}$ along the time axis and then recursively compute (21).

F. Time–Topology Alignment

In addition to $\mathcal{L}_{time}$, which learns instance-specific characteristics and temporal variation, we propose a contrastive objective for aligning time and topology modalities. Since the PD inherently captures topological features that are robust to perturbations, the proposed cross-modal alignment helps prevent potential information loss caused by augmentations. To this end, we first map the embedding vectors $r_{i,t}, r'_{i,t} \in \mathbb{R}^F$ from temporal encoder f_θ^{time}, and topological features $h_i \in \mathbb{R}^H$ to the multimodal latent space. Then, we apply projection headers $proj_{time}$ and $proj_{topo}$ as follows:

$$y_i = proj_{topo}\left(h_i\right) \tag{22}$$

$$z_i = proj_{time}\left(r_i^*\right) \tag{23}$$

$$z'_i = proj_{time}\left(r_i^{*\prime}\right)) \tag{24}$$

where r_i^* is an instance level representation of $r_{i,t}$ which is obtained by applying max pooling across all timestamps. The average of z_i and z'_i is computed as follows:

$$z_i^a = \frac{1}{2}\left(z_i + z'_i\right). \tag{25}$$

Next, we maximize the similarity between z_i^a and y_i, which are mapped into the multimodal latent space. The similarity between z_i^a and y_i is defined as $s(z_i^a, y_i) = (z_i^a \cdot y_i)/(\|z_i^a\|\|y_i\|)/\tau$, where τ refers to the temperature parameter. For a mini-batch B, (z_i^a, y_i) is regarded as a positive pair, whereas the remaining $2B - 2$ feature vectors constitute negative pairs. Our cross-modal contrastive loss between z_i^a and y_i can be formulated as follows:

$$\ell_{\text{align}}^{(i;time,topo)} = -\log \frac{\exp\left(s\left(z_i^a, y_i\right)\right)}{\sum_{j\in B}\left(\exp\left(s\left(z_i^a, y_j\right)\right) + \mathbb{1}_{[j\neq i]}\exp\left(s\left(z_i^a, z_j^a\right)\right)\right)}. \tag{26}$$

The final cross-modal contrastive loss is formulated as follows:

$$\mathcal{L}_{\text{cross}} = \frac{1}{2B}\sum_i\left(\ell_{\text{align}}^{(i;time,topo)} + \ell_{\text{align}}^{(i;topo,time)}\right). \tag{27}$$

In the time-to-topology direction $\ell_{\text{align}}^{(i;time,topo)}$, the temporal view z_i^a acts as the query, and the model is trained to align it with the matching topological representation y_i (target). Conversely, in the topology-to-time direction $\ell_{\text{align}}^{(i;topo,time)}$, the topological representation y_i becomes the query, and the temporal representation z_i^a is the target. This symmetric formulation enables the model to learn a mutual alignment between temporal and topological modalities. Leveraging the inherent robustness of PDs—which capture topological features resilient to perturbations—the proposed alignment further helps to prevent potential information loss introduced by data augmentations.

G. Overall Objective

The proposed model jointly optimizes CL in time modality and time–topology correspondence. The overall loss function can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{time} + \alpha\mathcal{L}_{cross} \tag{28}$$

where α is a hyperparameter.

Please note that instance discrimination using contrastive loss is not performed on the topology modality. Instead, we apply contrastive loss to the time modality and leverage topology modality to enhance time series representation learning. The assumption is that time–topology alignment allows learning comprehensive and complementary semantic information by incorporating temporal features and their corresponding topological features.

VI. Experiments

In this section, we conduct experiments on anomaly detection, classification, forecasting, and transfer learning tasks by applying TopoCL to state-of-the-art methods. Implementation details are presented in the Appendix.

A. Anomaly Detection

The anomaly detection is used for identifying and addressing unusual patterns or behaviors across various domains. In this task, Yahoo [34] and KPI [35] datasets are used. We apply TopoCL to TS2Vec [7], and the evaluation is performed under normal and cold-start settings. For the normal setting, each sample is divided into two portions: one for training and the other for testing. For the cold-start setting, the model is pretrained on the *FordA* dataset from the UCR archive before evaluating on each target dataset. Following the evaluation protocol proposed by [34], we determine whether the last point is an anomaly.

SPOT, DSPOT [36], DONUT [37], SR, and TS2vec methods are used as baselines for the normal setting, while FFT [38], Twitter-AD [39], and Luminol [40], SR, and TS2vec are used as baselines for the cold-start setting. As shown in Table I, in the normal setting, TopoCL improves the F_1 score by 1.9% and 2.2% on the Yahoo and KPI dataset, respectively. In the cold-start setting, our model outperforms the best baseline, achieving a 1.8% higher F_1 score on the Yahoo dataset and a 0.5% improvement on the KPI dataset.

B. Classification

In this section, we conduct a classification task to evaluate the performance of the proposed model using the UCR [41] and UEA [42] archives, selecting 125 univariate and 29 multivariate time-series datasets from each archive. We train an RBF kernel-based SVM classifier on top of the learned representation, following the evaluation protocol used in T-Loss [43]. The penalty factor for the RBF kernel is selected by performing a grid search on the validation set, exploring the range $\{10^i \mid i \in [-4, 4]\}$

We apply TopoCL to TS2vec and compare its performance with baselines including DTW, TST [44], TS-TCC [10], T-Loss [43], TNC [45], TS2vec, and InfoTS [24]. The classification results for these datasets are summarized in Table II. The classification accuracy of the proposed model outperforms the best baseline by 3.8% on the UCR datasets and by 1.4% on the UEA datasets. Detailed classification results are provided in the Appendix.

To further evaluate the effectiveness of TopoCL, we perform a significance test on the classification results. In Fig. 3, critical difference (CD) diagram [46] for Nemenyi tests on all datasets is presented. The CD diagram indicates that methods connected by a bold line do not have significantly different average ranks. From the results, we can conclude that the proposed model significantly outperforms other methods in average ranks.

TABLE I
Anomaly Detection Results of the Proposed Model and Baselines on Yahoo and KPI Datasets

	Yahoo			KPI		
	F_1	Prec.	Rec.	F_1	Prec.	Rec.
SPOT	0.338	0.269	0.454	0.217	0.786	0.126
DSPOT	0.316	0.241	0.458	0.521	0.623	0.447
DONUT	0.026	0.013	**0.825**	0.347	0.371	0.326
SR	0.563	0.451	0.747	0.622	0.647	**0.598**
TS2Vec	0.745	0.729	0.762	0.677	**0.929**	0.533
TS2Vec+TopoCL	**0.764**	**0.763**	0.764	**0.699**	0.924	0.562
Cold-start:						
FFT	0.291	0.202	0.517	0.538	0.478	0.615
Twitter-AD	0.245	0.166	0.462	0.330	0.411	0.276
Luminol	0.388	0.254	**0.818**	0.417	0.306	0.650
SR	0.529	0.404	0.765	0.666	0.637	**0.697**
TS2Vec	0.726	0.692	0.763	0.676	0.907	0.540
TS2Vec+TopoCL	**0.744**	**0.725**	0.764	**0.681**	**0.912**	0.543

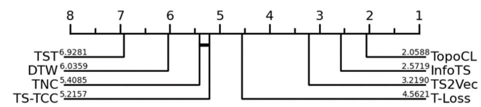


Fig. 3. CD diagram of representation learning methods on time-series classification tasks, using 125 UCR datasets and 29 UEA datasets.

C. Forecasting

In this section, we conduct two types of forecasting experiments to examine the plug-and-play capability of *TopoCL* across different self-supervised learning paradigms. Specifically, we apply *TopoCL* to: 1) the masked-modeling-based pretraining framework of PatchTST [47] and 2) contrastive-learning-based frameworks such as TS2Vec and CoST [9]. All models are evaluated using mean absolute error (MAE) and mean squared error (mse) as performance metrics.

TABLE II
CLASSIFICATION RESULTS OF THE PROPOSED MODEL AND BASELINES ON UCR AND UEA DATASETS

Methods		TS2vec+TopoCL	TS2Vec	InfoTS	T-Loss	TNC	TS-TCC	TST	DTW
29 UEA datasets	Avg. Acc.	**0.748 (+2.6%)**	0.712	0.722	0.675	0.677	0.682	0.635	0.650
	Avg. Rank	**2.293**	3.948	3.293	4.638	5.534	5.017	5.983	5.293
125 UCR datasets	Avg. Acc.	**0.853 (+1.5%)**	0.838	0.838	0.806	0.761	0.757	0.641	0.727
	Avg. Rank	**2.004**	3.048	2.392	4.540	5.376	5.268	7.156	6.216

TABLE III
MULTIVARIATE FORECASTING RESULTS WHEN TOPOCL INTEGRATED WITH PATCHTST

Models		TopoCL		ASAP		iTransformer		PatchTST		Crossformer		TimesNet		DLinear		MRformer		NonStat		SCINet	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.374**	**0.400**	0.393	0.407	0.386	0.405	0.406	0.413	0.423	0.448	0.384	0.402	0.386	**0.400**	0.832	0.718	0.513	0.491	0.654	0.599
	192	**0.424**	0.431	0.445	0.433	0.441	0.436	0.458	0.441	0.471	0.474	0.436	**0.429**	0.437	0.432	0.884	0.738	0.534	0.504	0.719	0.631
	336	**0.463**	**0.450**	0.483	0.449	0.487	0.458	0.497	0.464	0.570	0.546	0.491	0.469	0.481	0.459	0.968	0.777	0.588	0.535	0.778	0.659
	720	0.505	0.485	0.478	**0.469**	0.503	0.491	0.498	0.483	0.653	0.621	0.521	0.500	0.519	0.516	1.019	0.972	0.643	0.616	0.836	0.699
ETTh2	96	**0.281**	**0.337**	0.294	0.342	0.297	0.349	0.300	0.347	0.745	0.584	0.340	0.374	0.333	0.387	1.213	0.911	0.476	0.458	0.707	0.621
	192	**0.353**	**0.383**	0.377	0,393	0.381	0.409	0.384	0.396	0.877	0.656	0.402	0.414	0.477	0.476	2.916	1.363	0.512	0.493	0.860	0.689
	336	**0.364**	**0.400**	0.381	0.409	0.428	0.432	0.393	0.403	1.043	0.731	0.452	0.452	0.594	0.541	3.264	1.501	0.552	0.551	1.000	0.744
	720	0.413	0.435	**0.411**	**0.434**	0.427	0.455	0.425	0.437	1.104	0.763	0.462	0.468	0.831	0.657	3.551	1.584	0.562	0.560	1.249	0.838
ETTm1	96	**0.316**	**0.356**	0.317	0.357	0.334	0.368	0.328	0.367	0.404	0.426	0.338	0.375	0.345	0.372	0.547	0.539	0.386	0.398	0.418	0.438
	192	**0.358**	**0.383**	0.359	0.382	0.377	0.391	0.367	0.388	0.450	0.451	0.374	0.387	0.380	0.389	0.629	0.583	0.459	0.444	0.439	0.450
	336	**0.383**	**0.403**	0.388	0.405	0.426	0.420	0.398	0.407	0.532	0.515	0.410	0.411	0.413	0.413	1.072	0.789	0.495	0.464	0.490	0.485
	720	**0.446**	**0.441**	0.448	0.437	0.491	0.459	0.454	0.439	0.666	0.589	0.478	0.450	0.474	0.453	1.012	0.776	0.585	0.516	0.595	0.550
Avg.		**0.390**	**0.408**	0.398	0.411	0.414	0.423	0.409	0.415	0.661	0.567	0.424	0.428	0.473	0.458	1.492	0.938	0.525	0.5025	0.729	0.617

1) TopoCL With PatchTST: In this task, the PatchTST architecture is employed as the encoder and trained in a self-supervised manner using a masked-modeling objective. The model is first pretrained on the ETT [48] dataset and subsequently finetuned on the same data for task-specific long-term forecasting. TopoCL is integrated into this training pipeline to enhance representation learning during the pretraining stage. The experiments focus on long-term time-series forecasting with prediction horizons of 96, 192, 336, and 720 time steps.

We compare our method against ten state-of-the-art time-series forecasting baselines including ASAP [49], iTransformer [50], PatchTST, Crossformer [51], MRformer [52], NonStat [53], SCINet [54], TimesNet [55], and DLinear [56]. To ensure a fair comparison, the input sequence length for all models is fixed at 96. As summarized in Table III, TopoCL consistently outperforms end-to-end forecasting baselines that directly optimize the prediction loss. It achieves improvements across most prediction horizons on multivariate long-term forecasting benchmarks, with an average reduction of 2.0% in mse and 0.7% in MAE over the best-performing baseline.

2) TopoCL With TS2Vec and CoST: To further assess the plug-and-play capability of *TopoCL* under CL paradigms, we integrate it into two representative self-supervised frameworks: TS2Vec and CoST [9]. Following [7] and [9], a ridge regressor with L_2 regularization is trained on top of the final time-step representation r_t to forecast future values. Specifically, the model predicts the next 24, 48, 168, and 720 time steps based on the preceding observations. The regularization coefficient is selected using the validation set, searching over $\{0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000\}$.

We use Informer [48], LogTrans [57], TCN [58], LSTnet [59], TS2Vec, and CoST as baselines. The evaluation results for multivariate forecasting are shown in Table IV. In general, CoST + TopoCL outperforms the baselines in most cases, achieving reductions of 1.76% in multivariate forecasting, in terms of mse. Furthermore, TS2Vec + TopoCL demonstrates a 3.33% decrease in average mse compared with TS2Vec alone for multivariate forecasting.

D. Transfer Learning

Transfer learning on time-series data brings a unique challenge, stemming from the complex nature of the data, such as rapidly changing trends and shifts in temporal dynamics. In this section, we conduct transfer learning, following the experimental settings proposed by TF-C [26]. Specifically, the model is pretrained on the SleepEEG dataset and subsequently finetuned on the Epilepsy [60], Gesture [61], FD-B [62], and EMG [63] datasets.

To assess the generality of the proposed approach, TopoCL is applied to two distinct self-supervised learning paradigms: CL (TS-TCC [10]) and masked modeling (SimMTM [25]). The method is compared with several state-of-the-art baselines, including TS-TCC, SimMTM, TS2Vec, CoST, Ti-MAE [64], TF-C, and TST. Performance is evaluated using accuracy and F_1 score. As shown in Table V, both SimMTM and TS-TCC exhibit performance improvements when combined with

TABLE IV

MULTIVARIATE FORECASTING RESULTS WHEN TOPOCL INTEGRATED WITH COST AND TS2VEC

Models		CoST+		CoST		TS2Vec+		TS2vec		Informer		LogTrans		TCN		LSTnet	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	24	**0.383**	**0.428**	0.384	**0.428**	0.560	0.521	0.588	0.531	0.577	0.549	0.686	0.604	0.583	0.547	1.293	0.901
	48	**0.432**	**0.459**	0.437	0.464	0.597	0.544	0.624	0.556	0.685	0.625	0.766	0.757	0.670	0.606	1.456	0.960
	168	**0.637**	**0.580**	0.643	0.582	0.734	0.631	0.764	0.639	0.931	0.752	1.002	0.846	0.811	0.680	1.997	1.214
	336	**0.793**	**0.671**	0.812	0.679	0.896	0.717	0.937	0.731	1.215	0.896	1.397	1.291	1.132	0.815	2.143	1.380
	720	**0.916**	**0.754**	0.970	0.771	1.054	0.799	1.066	0.800	1.215	0.896	1.397	1.291	1.165	0.813	2.143	1.380
ETTh2	24	0.436	0.498	0.463	0.512	**0.413**	**0.482**	0.423	0.489	0.720	0.665	0.828	0.750	0.935	0.754	2.742	1.457
	48	0.681	0.633	0.711	0.648	**0.618**	0.607	0.619	**0.605**	1.457	1.001	1.806	1.034	1.300	0.911	3.567	1.687
	168	**1.561**	**0.986**	1.583	0.997	1.793	1.064	1.845	1.074	3.489	1.515	4.070	1.681	4.017	1.579	3.242	2.513
	336	**1.795**	**1.062**	1.809	1.064	2.136	1.182	2.194	2.197	2.723	1.340	3.875	1.763	3.460	1.456	2.544	2.591
	720	**1.982**	**1.105**	1.989	1.100	2.480	1.319	2.636	1.370	3.467	1.473	3.913	1.552	3.106	1.381	4.625	3.709
ETTm1	24	**0.244**	**0.327**	0.246	0.330	0.438	0.435	0.448	0.440	0.323	0.369	0.419	0.412	0.363	0.397	1.968	1.170
	48	**0.329**	**0.384**	0.334	0.389	0.574	0.513	0.595	0.524	0.494	0.503	0.507	0.583	0.542	0.508	1.999	1.215
	96	**0.376**	**0.417**	0.382	0.423	0.613	0.543	0.629	0.551	0.678	0.614	0.768	0.792	0.666	0.578	2.762	1.542
	288	**0.470**	**0.484**	0.477	0.489	0.673	0.585	0.690	0.594	1.056	0.786	1.462	1.320	0.991	0.735	1.257	2.076
	672	**0.621**	**0.574**	0.625	0.576	0.766	0.644	0.780	0.652	1.192	0.926	1.669	1.461	1.032	0.756	1.917	2.941
Avg.		**0.777**	**0.624**	0.791	0.630	0.956	0.706	0.989	0.784	1.348	0.861	1.638	1.076	1.385	0.834	2.377	1.782

TABLE V

TRANSFER LEARNING RESULTS

	Epilepsy		FD-B		Gesture		EMG	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
TS2Vec	0.8395	0.9045	0.4790	0.4389	0.6917	0.6570	0.7854	0.6766
CoST	0.8840	0.7688	0.4706	0.3479	0.6833	0.6642	0.5365	0.3527
Ti-MAE	0.7345	0.7720	0.6798	0.6336	0.7554	0.6932	0.6352	0.5832
TF-C	0.9495	0.9149	0.6938	0.7487	0.7642	0.7572	0.8171	0.7683
TST	0.8021	0.4011	0.4640	0.4134	0.6917	0.6601	0.4634	0.2111
TS-TCC	0.9253	0.8633	0.5499	0.5418	0.7188	0.6984	0.7889	0.5904
SimMTM	0.9481	0.9123	0.6291	0.6867	0.7833	0.7594	0.9512	0.9362
TS-TCC+	0.9483	0.9133	**0.7621**	**0.8174**	0.7667	0.7605	**1.0000**	**1.0000**
SimMTM+	**0.9525**	**0.9209**	0.6727	0.7324	**0.8083**	**0.7851**	0.9756	0.9813

TopoCL. In particular, integrating TopoCL with TS-TCC leads to a notable performance gain.

VII. ANALYSIS

A. Ablation Study

In this section, we compared TopoCL to its five variants on 128 UCR datasets to study the effectiveness of the proposed components. The five variants are as follows: *1) w/o time–topology alignment:* removes the time–topology cross-modal alignment from TopoCL; *2) w/o time domain contrastive loss* removes the instance and temporal contrast from f_θ^{topo}; *3) w/o* H_0 excludes the 0-D persistent homology when computing the PD, *4) w/o* H_1 excludes the 1-D persistent homology; and *5) avg-pool* replaces the symmetric aggregate function (max-pool) in f_θ^{topo} with avg-pool. As shown in Table VI, any absence of the components leads to a decrease in performance, showing that all components are imperative.

B. Runtime and Memory Analysis

In this section, the runtime and memory consumption of the proposed framework are evaluated using two representative datasets: the *EigenWorms* dataset, which contains the longest multivariate time series (six channels and 17 984 time steps), and the *DuckDuckGeese* dataset, which has the highest dimensionality (1345 channels and 60 time steps).

Because computational and memory costs increase with the number of persistence points generated during PD computation, this quantity is reported for both datasets. The maximal number of persistence points reached 520 910 for *DuckDuckGeese* and 33 021 for *EigenWorms*. Although persistent homology computations are performed entirely as an offline preprocessing step, the large volume of topological features caused *DuckDuckGeese* to exceed the memory capacity of an NVIDIA RTX 3090 GPU (24-GB VRAM). To mitigate this issue, farthest point sampling (FPS) was applied to the point cloud $x_i^p \in X_{\text{topo}}$, reducing redundancy while preserving the underlying geometric structure. The impact of FPS was examined by retaining 10k and 1k points, respectively, and the corresponding runtime, GPU memory usage, and classification accuracy are summarized in Table VII.

TABLE VI

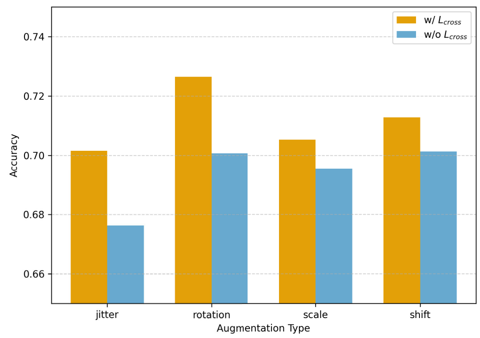
ABLATION RESULTS

	Avg. Accuracy
TopoCL	**0.852**
w/o time-topology alignment	0.828 (-2.4%)
w/o time domain contrastive loss	0.804 (-4.8%)
w/o H_0	0.829 (-2.3%)
w/o H_1	0.833 (-1.9%)
Avg-pool	0.845 (-0.7%)

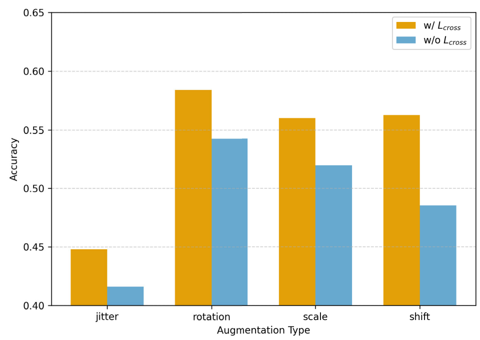
TABLE VII

EFFECT OF FPS DOWNSAMPLING ON RUNTIME, MEMORY USAGE, AND ACCURACY

Metric	EigenWorms			DuckDuckGeese		
	Full	10k	1K	Full	10K	1K
Iteration time (s)	0.31	0.21	0.19	-	0.08	0.07
Max Peak (MB)	6829.24	5760.56	4234.56	-	1172.23	237.15
Accuracy	0.91	0.91	0.88	-	0.56	0.55



(a)



(b)

Fig. 4. Comparison of classification accuracy under different augmentation techniques with and without $\mathcal{L}_{cross}$) on (a) crop and (b) RefrigerationDevices datasets.

As shown in the results, applying FPS to our framework reduces memory consumption, while causing only a marginal drop in accuracy. These results demonstrate that although the large volume of topological features can lead to computational overhead, such challenges can be effectively mitigated through sampling strategies.

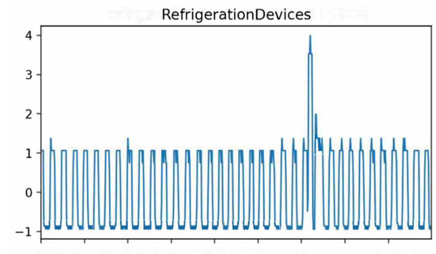
C. Robustness Under Data Augmentation

The robustness of TopoCL was evaluated against different data augmentation techniques using the *Crop* and *RefrigerationDevices* datasets from the UCR archive. Specifically, common augmentation methods, including jittering, scaling, shifting, and rotation, were applied individually.

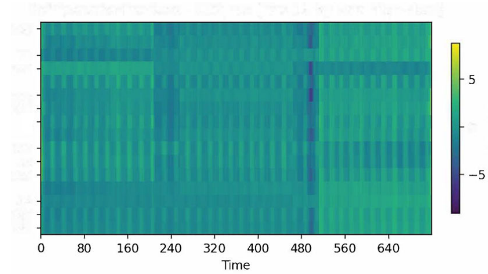
The classification accuracy was measured for both configurations—with and without the proposed topological regularization—and the results are summarized in Fig. 4. While the performance gain from each augmentation varies across datasets, incorporating $\mathcal{L}_{\text{cross}}$ generally led to improved accuracy under different augmentation settings. These results empirically demonstrate that time–topology alignment can help compensate for the information loss introduced during data augmentation.

D. Visualized Explanation

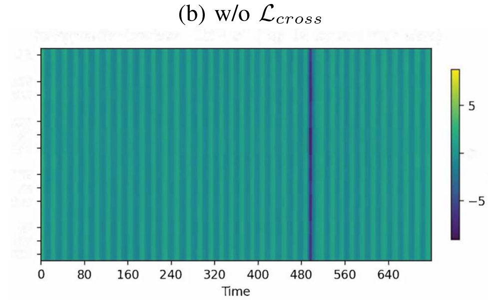
In this section, we visualize the learned embeddings obtained from the time–topology alignment. The analysis is conducted on the *RefrigerationDevices* dataset, which primarily exhibits periodic behavior with occasional sharp transient peaks. The first test sample is selected as a representative case for evaluation.



(a) Raw Time series



(b) w/o $\mathcal{L}_{cross}$



(c) Full model

Fig. 5. Visualized explannations. (a) Raw time series. (b) Learned embeddings of TopoCL w/o $\mathcal{L}_{cross}$. (c) Learned embedding of TopoCL.

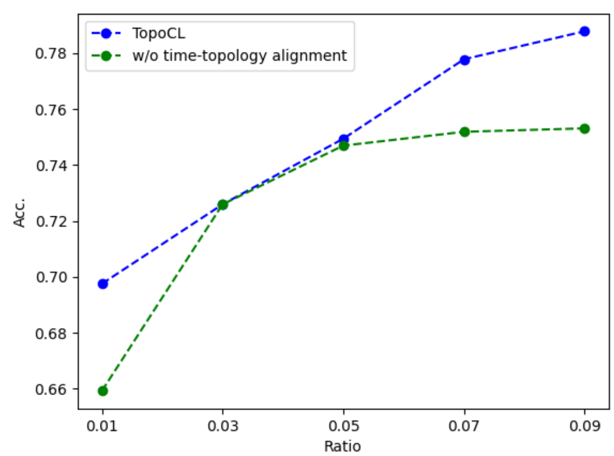


Fig. 6. Classification results in a limited data sample scenario.

To illustrate the representative structure of the learned embeddings (see Fig. 5), we focus on the 16 latent dimensions with the highest variance. Without the time–topology correspondence term $\mathcal{L}$cross, the representation shows irregular fluctuations and diffuse responses around the transient anomaly ($t \approx 480$), suggesting that the model mainly captures local amplitude variations. In contrast, incorporating $\mathcal{L}$cross produces more periodic and structured embeddings, with the transient anomaly localized to only a few dimensions.

E. Limited Data Scenario

Due to sensor malfunctions, data transmission errors, or manual data entry issues, a sufficient amount of data is not always guaranteed. To verify the effectiveness of the proposed method in scenarios with limited data, we adjusted the proportion of training samples. *FordB* dataset from the UCR archive is chosen for this analysis. The classification results using only 1%–9% of the original training dataset are shown in Fig. 6. The proposed method consistently outperforms *w/o time–topology alignment*. This result suggests that leveraging topological properties can capture the intrinsic characteristics of the data, even with a small amount of data.

VIII. Conclusion

This article presents TopoCL, a topological CL for time series. We treat temporal and topological properties of time series as distinct modalities and propose a joint learning objective to enhance understanding of both, while also improving robustness to data augmentations. TopoCL is evaluated across multiple tasks: time-series classification, forecasting, anomaly detection, and transfer learning. The experimental results demonstrate the universality, generalization capability, and effectiveness of the proposed model. Our ablations show that the joint learning of time modality contrastive loss and time–topology correspondences enhances these capabilities. Moreover, TopoCL shows consistent performance across various data augmentation techniques, suggesting its effectiveness in mitigating information loss resulting from these augmentations. In the future, we aim to extend TopoCL to accommodate large scale and diverse pretraining datasets, thereby advancing toward a foundation model for time-series analysis.

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During the preparation of this work they used ChatGPT and Grammarly in order to refine the writing and check the grammar.

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